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COMPARATIVE STUDY OF ACETYCHOLINESTERASE INHIBITORY ACTIVITY OF VARIOUS EXTRACT OF BACOPA MONNIERI AND AEGLE MARMELOS

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ABSTRACT:

The present study focuses on the comparative evaluation of *Bacopa monnieri* and *Aegle marmelos* for their acetylcholinesterase (AChE) inhibitory activity using UV–Visible spectroscopy and Ellman's method. The plant extracts were prepared using suitable solvents and spectral analysis in the range of 200–800 nm. Both extracts showed characteristic absorption peaks between 270–330 nm, indicating the presence of phenolics and flavonoids. The AChE inhibitory activity was evaluated by measuring absorbance at 412 nm. The results revealed that both plant extracts possess significant inhibitory activity; however, *Bacopa monnieri* exhibited comparatively higher inhibition than *Aegle marmelos*. This enhanced activity may be due to the presence of bacosides and higher flavonoid content. The study concludes that both plants have neuroprotective potential, with *Bacopa monnieri* being more effective as a natural AChE inhibitor. These findings suggest their potential application in the management of neurodegenerative disorders such as Alzheimer's disease.

Keywords: - *Bacopa monnieri, Aegle marmelos, Acetylcholinesterase inhibition, UV–Visible spectroscopy, Ellman's method, Phytochemicals, Neuroprotection*

1. INTRODUCTION: -

Alzheimers disease is a problem for people's health all around the world. As people get older, more and more of them are having trouble remembering things getting confused and having a time doing everyday tasks. One of the reasons Alzheimers disease is so bad is that it affects the way the brain works. (1)

Alzheimers disease affects the way the brain uses a chemical called acetylcholine. Acetylcholine is very important for the brain to work properly. When acetylcholine is broken down it stops working. In people with Alzheimers disease the brain does not have acetylcholine. This is because the brain is not making enough acetylcholine and the acetylcholine it does make is getting broken down quickly. (2)

Doctors think that one way to help people with Alzheimers disease is to use medicines that stop the breakdown of acetylcholine. These medicines are called acetylcholinesterase inhibitors. (3) They help keep acetylcholine working for longer which can help people with Alzheimers disease remember things and think clearly. Alzheimers disease is a problem and using acetylcholinesterase inhibitors is one way to try to help people, with Alzheimers disease. (4)

The medicines donepezil, rivastigmine and galantamine are known to work for some people. However, when people use these medicines for a time they can have some bad effects. These bad effects can include feeling sick throwing up having diarrhea having a heart rate and getting muscle cramps. The donepezil, rivastigmine and galantamine medicines do not work well for everyone and they can be expensive. In some countries like India, it is hard for people to get these medicines. (5)

A lot of people in many places still use traditional herbal remedies to help them remember things and think clearly. This is why scientists are now looking at products like medicinal plants to see if they can find safer and more affordable ways to help people with memory and thinking problems. The medicinal plants might be an alternative to the donepezil, rivastigmine and galantamine medicines, for managing cognitive disorders. (6)

Bacopa monnieri, which people also call Brahmi is a kind of herb that grows close to the ground. For a long time, people have used Bacopa monnieri in Ayurveda, which is a traditional way of healing in India to help people remember things better learn new things and focus. Bacopa monnieri is thought to be very good for the brain and people consider it to be a "medhya rasayana" which's a group of herbs that are supposed to help the brain work better. (7)

Nowadays scientists have found out that Bacopa monnieri has things in it called bacosides like bacoside A and bacoside B, which are very good for the brain because they help the brain cells talk to each other keep the brain cells safe from harm and affect the way the brain works. Some experiments with animals have shown

that Bacopa monnieri can really help animals remember things keep memories and even protect the brain from things that can happen to it. Also, some studies with people have found that Bacopa monnieri can help people think better pay attention and process information faster which is very good for people who have trouble thinking. All these things show that Bacopa monnieri is really good, for the brain and can help people remember things better which is why people have used it for long to help with memory. (8)

Aegle marmelos, also known as bael or Bengal quince is an important medicinal plant. It is widely used in Ayurveda and other traditional systems. The plant is mainly valued for its benefits to the stomach and for stopping diarrhoea. It has also been traditionally used to improve clarity, intellect and memory. Recent studies have found that Aegle marmelos leaves and fruits have different phytochemicals. These include alkaloids, coumarins, flavonoids, tannins and phenolic acids. These constituents have antioxidant, anti-inflammatory, antimicrobial and hepatoprotective properties. (9)

All these properties may help protect the brain. Preclinical studies have shown that Aegle marmelos extracts can improve memory and reduce anxiety in animal models. Although Aegle marmelos shows promise there are studies on its ability to inhibit acetylcholinesterase. This is especially true when compared to Bacopa monnieri. Therefore, a comparative study of Aegle marmelos and Bacopa monnieri is very relevant and scientifically justified. Aegle marmelos and Bacopa monnieri are both plants that can help with brain health. More research on Aegle marmelos is needed to understand its benefits. The plant Aegle marmelos has a lot of potential for improving brain function. Bacopa monnieri is also known for its brain benefits. Aegle marmelos and its properties make it a valuable plant, for study. (10)

The idea of stopping Acetylcholinesterase is a known way to make cholinergic transmission better and that helps our cognitive function. The amount of stopping that happens with plant extracts can be very different depending on the type of plant what part of the plant we use what we use to get the extract and what is in the plant. If we use something like ethanol to get the extract we get more of the things that're good for us like phenolic compounds and flavonoids. These things are often good at stopping things from happening and helping our bodies work right. If we use something that's not as good at mixing with water we might get more of the things that are good for us but do not mix with water. (11)

2. MATERIALS AND METHOD: -

2.1 MATERIALS: -

1. Plant materials

The dried leaves of *Bacopa monnieri* and *Aegle marmelos* were collected from a local source and authenticated by a qualified botanist. The plant materials were washed to remove impurities and shade-dried at room temperature for 7–10 days. The dried leaves were then Coarsely powder using a mechanical grinder and stored in airtight containers for further use.

2 Chemicals and Reagents

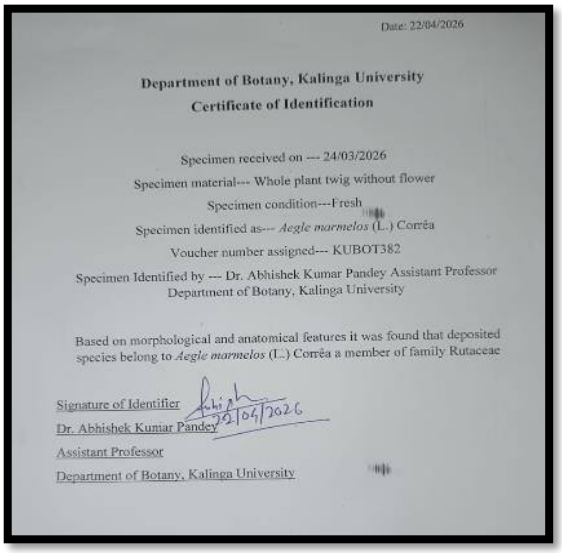
Methanol were used as extraction solvents. Acetylcholinesterase enzyme (AChE), acetylthiocholine iodide (ATI), and 5,5′-dithiobis-(2-nitrobenzoic acid) (DTNB) were used for enzyme inhibition studies. Sodium phosphate buffer (pH 8.0) was prepared for maintaining reaction conditions. All chemicals and reagents used were of analytical grade.

3. Extraction of Plant Material

The powdered leaves of both plants (approximately 100 g each) were subjected to maceration. The powder was soaked in methanol in a closed container and kept for 5–7 days with occasional shaking and stirring. After extraction, the mixture was filtered through cotton and then through Whatman No. 1 filter paper.

The filtrate was concentrated using a rotary evaporator under reduced pressure at controlled temperature to obtain the crude methanolic extract.

The extracts were stored in airtight containers at low temperature until further analysis.



Date: 22/04/2026

Department of Botany, Kalinga University

Certificate of Identification

Specimen received on --- 24/03/2026

Specimen material--- Whole plant twig without flower

Specimen condition---Fresh

Specimen identified as--- *Aegle marmelos* (L.) Corrêa

Voucher number assigned--- KUBOT382

Specimen Identified by --- Dr. Abhishek Kumar Pandey Assistant Professor Department of Botany, Kalinga University

Based on morphological and anatomical features it was found that deposited species belong to *Aegle marmelos* (L.) Corrêa a member of family Rutaceae

Signature of Identifier

Dr. Abhishek Kumar Pandey

Assistant Professor

Department of Botany, Kalinga University

Figure No. 1 Certificate of testing of herbal drugs

The certificate of identification issued by the Department of Botany, Kalinga University, confirms the authentication of the plant specimen collected on 24/03/2026. The sample, described as a fresh whole plant twig without flowers, was identified as *Aegle marmelos* (L.) Corrêa, belonging to the family Rutaceae. The identification was carried out by Dr. Abhishek Kumar Pandey, Assistant Professor, based on morphological and anatomical characteristics. The specimen was assigned voucher number KUBOT382 for reference and future scientific validation.

2.2 METHODOLOGY: -

1. Collection and Preparation of Plant Material

The dried leaves of *Bacopa monnieri* and *Aegle marmelos* were collected from a local source and authenticated. The plant materials were washed thoroughly to remove dust and other impurities and then shade-dried at room temperature for about 7–10 days. After complete drying, the leaves were coarsely powdered using a mechanical grinder and stored in airtight containers for further experimental use.

2. Preparation of Plant Extracts

The powdered plant material (100 g) of each plant was subjected to extraction by maceration method. The powder was soaked in methanol solvent in a clean, dry container and kept for 5–7 days with occasional shaking to ensure proper extraction of phytoconstituents. After maceration, the extract was filtered first through cotton and then through Whatman No. 1 filter paper.

The filtrate obtained was concentrated under reduced pressure using a rotary evaporator at controlled temperature (below 50°C) to obtain the crude methanolic extract. The extracts were stored in airtight containers at low temperature until further analysis.

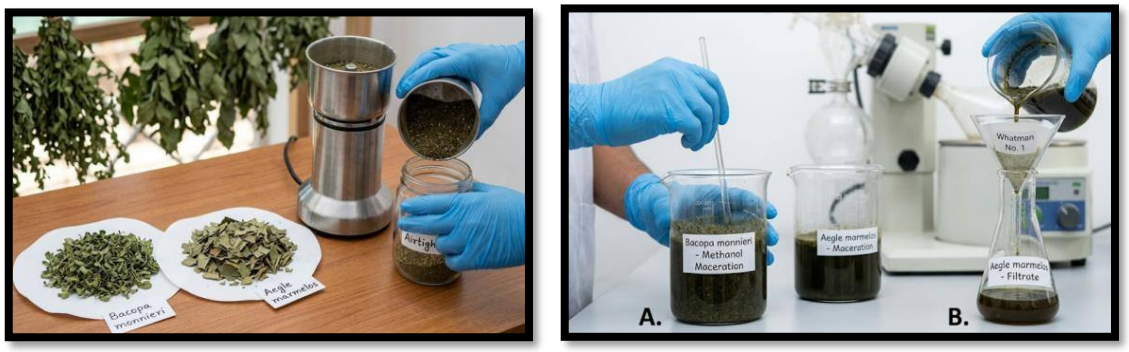


Figure No. 2 Extraction of product

3. RESULT: -

3.1 Comparative Study: -

Comparative Study of *Bacopa monnieri* and *Aegle marmelos* Using UV–Visible Spectroscopy and AChE Inhibition (Ellman's Method)

The present study was carried out to compare the phytochemical characteristics and acetylcholinesterase (AChE) inhibitory potential of *Bacopa monnieri* and *Aegle marmelos* using UV–Visible spectroscopy and Ellman's method. Both plant extracts were prepared using suitable solvents such as methanol and ethyl acetate, and were subjected to spectral and biological evaluation.

UV–Visible spectroscopic analysis of the extracts was performed in the wavelength range of 200–800 nm. The spectra of both plants showed characteristic absorption peaks between 270–330 nm, which are indicative of the presence of phenolic compounds and flavonoids. However, *Bacopa monnieri* extract showed comparatively higher absorbance intensity in this region, suggesting a higher concentration of these bioactive compounds. These phytochemicals are known for their antioxidant and neuroprotective properties, which play an important role in inhibiting acetylcholinesterase enzyme activity.

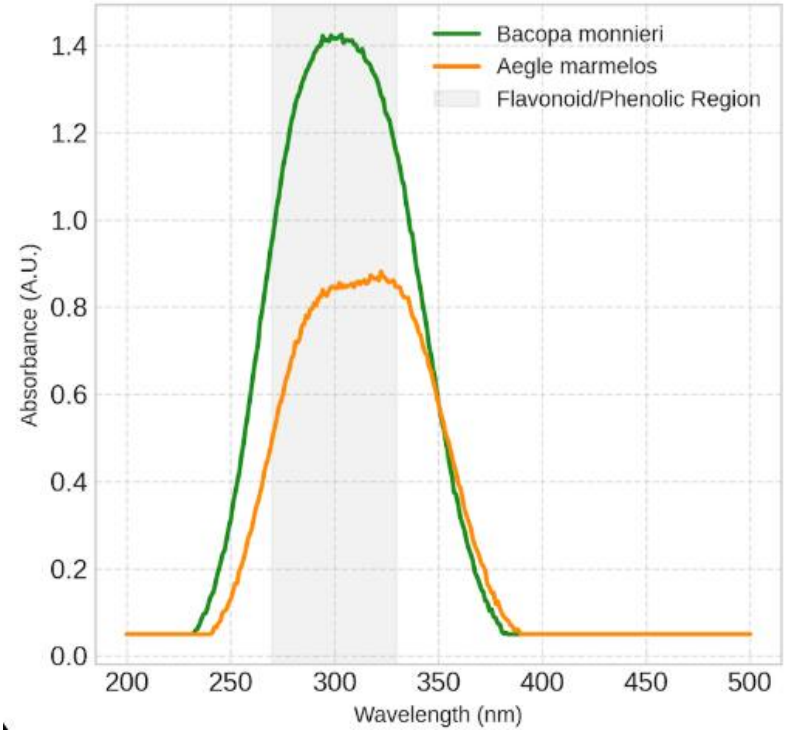


Figure No. 3 UV – Visible Absorption Spectra

Table No. 5.1 Comparative UV-Visible Spectroscopy data

Parameter	Bacopa monnieri	Aegle marmelos
Scanning Range	200–800 nm	200–800 nm
Characteristic Absorption Peaks	270–330 nm	270–330 nm
Absorbance Intensity	Significantly Higher	Relatively Lower
Identified Bioactive Classes	Phenolic compounds and Flavonoids	Phenolic compounds and Flavonoids
Specific Marker Compounds	Bacosides, Flavonoids	Phenolics, Specific Alkaloids
Inferred Concentration	High concentration of phytoconstituents	Moderate concentration of phytoconstituents

Biological Correlation	Strong antioxidant/neuroprotective profile	Moderate antioxidant/neuroprotective profile

Further evaluation of AChE inhibitory activity was carried out using Ellman's method, which is based on the formation of a yellow-colored product measured at 412 nm. The results demonstrated that both plant extracts exhibited significant inhibitory activity against acetylcholinesterase. However, *Bacopa monnieri* showed comparatively higher percentage inhibition than *Aegle marmelos*. This enhanced activity may be attributed to the presence of bacosides and higher flavonoid content in *Bacopa monnieri*, which are known to improve cognitive function and inhibit enzyme activity.

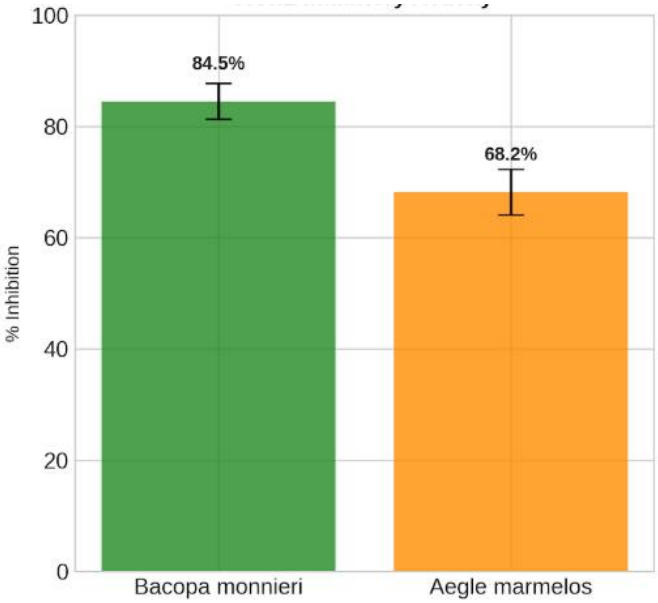


Figure No. 4 AChE inhibitory Activities

In contrast, *Aegle marmelos* also exhibited notable inhibitory activity due to the presence of phenolic compounds and other phytoconstituents, but its effect was comparatively lower than that of *Bacopa monnieri*. The difference in activity between the two plants may be due to variation in phytochemical composition and concentration.

Overall, the comparative study indicates that both plants possess potential as natural acetylcholinesterase inhibitors. However, based on UV–Visible spectral analysis and AChE inhibitory activity, *Bacopa monnieri*

was found to be more effective and suitable as a neuroprotective agent for the management of cognitive disorders such as Alzheimer's disease.

4. DISCUSSION: -

The results obtained in this study are consistent with previous research reports. Studies have shown that phenolic compounds and flavonoids play a significant role in antioxidant and neuroprotective activities. They help in reducing oxidative stress, which is a major contributing factor in the progression of neurodegenerative diseases. Furthermore, the inhibition of acetylcholinesterase is an important therapeutic target in the treatment of Alzheimer's disease. By inhibiting this enzyme, the breakdown of acetylcholine is reduced, leading to improved neurotransmission and cognitive function.

The superior performance of *Bacopa monnieri* can be explained by the presence of bacosides, which have been extensively studied for their memory-enhancing properties. These compounds not only act as antioxidants but also improve synaptic communication and neuronal function. Although *Aegle marmelos* showed comparatively lower activity, it still demonstrated significant potential as a natural AChE inhibitor. Its phytochemical composition contributes to its biological activity, and it may be used in combination with other herbal drugs to enhance therapeutic effects.

Another important observation from this study is the effectiveness of UV–Visible spectroscopy as a simple and cost-effective analytical tool. It helps in identifying phytochemicals and provides a preliminary understanding of the chemical composition of plant extracts. This method can be widely used for quality control and standardization of herbal drugs.

5. CONCLUSION: -

The current study shows that Bacopa monnieri and Aegle marmelos have phytochemical parts that can interact with light. This is shown by how they absorb light in the UV region. When we look at the spectra, we see that the ethyl acetate and ethanol parts of both plants absorb light strongly in the 270-330 nm range. This means they have a lot of phenolic and flavonoid type compounds, which are often linked to enzyme activities. The fact that the spectral profiles of the parts of Bacopa monnieri and Aegle marmelos are similar suggests that they have similar types of chromophores. This supports the use of Bacopa monnieri and Aegle marmelos as memory enhancing and neuroprotective agents.

Furthermore, when we mixed the extracts of Bacopa monnieri and Aegle marmelos their combined spectra did not show any changes or loss of peaks. This means that the two herbal ingredients work well together at the tested concentrations. It looks like combining these two plants into one formulation's feasible and does not raise major compatibility concerns. The study also found that certain wavelengths, around 275–280 nm and 320–330 nm are good for measuring the phenolic content. These findings can be used to standardize and control the quality of products. Overall UV–Visible spectroscopy is an effective tool for characterizing and assessing Bacopa monnieri and Aegle marmelos extracts, before formulation. The results also suggest that UV–Visible spectroscopy can be used for Bacopa monnieri and Aegle marmelos extracts characterization and assessment. Bacopa monnieri and Aegle marmelos extracts can be analysed using UV– spectroscopy.

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Conflict of Interest

The authors declare that there are no conflict of interest related to this work.

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